AfriEconQA: A Benchmark for Quantitative and Temporal Reasoning over World Bank Economic Reports

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Abstract

Reliable question answering over long institutional documents requires more than topical retrieval: a system must localize the exact passage that supports a claim and preserve precise numerical and temporal detail when the same indicator recurs across years, countries, and projection horizons in dense, repetitive prose. Existing question-answering benchmarks rarely test this combination of exact grounding and temporal precision at document scale. We introduce **AfriEconQA**, a benchmark for document-grounded question answering built from **220** World Bank economic reports on African economies, a corpus whose claims are tied to specific fiscal periods, countries, and projection states. AfriEconQA contains **4,309** evidence-linked QA instances across five reasoning categories: Factoid (1,093), List (879), Multiple Choice (944), Synthesis (710), and Comparison (683), targeting quantitative extraction, set recovery, discrimination, causal integration, and cross-period or cross-country reasoning. Each instance carries supporting evidence and source provenance and is constructed through an agentic generation pipeline with evidence-grounding checks and a stratified human-validation subset for gold-label auditing. We evaluate Qwen 3.6 35B, DeepSeek v4-pro, and Gemma 4 12B IT under zero-shot, oracle, and hybrid retrieval-augmented generation (RAG) conditions on the held-out test split ($n = 862$). Across all three models, retrieval yields substantial gains, yet the best RAG system reaches only 0.545 F1, with residual errors concentrated in list extraction, synthesis, and temporally scoped comparison. AfriEconQA therefore poses a hard open challenge for exact numerical and temporal grounding over long institutional economic reports.

1 Introduction

In document-grounded question answering, a topically relevant passage may still support the wrong answer. Long institutional economic documents often repeat the same indicator across countries, fiscal periods, report editions, and forecast horizons. A system can therefore retrieve the correct subject while selecting a historical value instead of a projection, a regional aggregate instead of a country estimate, or a policy target instead of an observed outcome. These errors alter the substantive meaning of an answer while leaving it fluent and numerically plausible. Reliable QA in this setting requires exact evidence localization together with numerical and temporal fidelity.

General advances in language-model capability do not by themselves resolve this problem Ni et al. (2025), particularly when answers must be recovered from a specialized document collection rather than parametric memory Daull et al. (2025). The closest financial QA benchmarks, including FinQA Chen et al. (2021), DocFinQA Reddy et al. (2024), and FinanceBench Islam et al. (2023), evaluate reasoning over corporate reports, financial statements, and company filings. DocFinQA stresses long-context retrieval over SEC filings. FinanceBench targets open-book corporate filing QA, while SEC-QA Lai et al. (2025) emphasizes metric-driven multi-document financial questions. None of these settings directly target institutional development reports, which combine narrative policy analysis with

repeated statistics, historical outcomes, current estimates, and forward projections. Consequently, it remains difficult to measure whether QA systems can locate and use the exact country-, period-, and projection-scoped evidence needed for macroeconomic analysis.

We introduce **AfriEconQA** to evaluate this capability. The benchmark contains **4,309** evidence-linked QA instances constructed from **220** World Bank economic reports focused on African economies. Each instance includes a question, answer, supporting evidence, and source metadata. Five question categories expose complementary capabilities: Factoid questions target exact values or entities, while List questions require complete set recovery. Multiple Choice questions test discrimination among plausible alternatives, Synthesis questions integrate connected policy claims, and Comparison questions reason across periods, countries, indicators, or projection states.

AfriEconQA uses an agentic pipeline to generate candidate questions, followed by evidence-grounding checks. Candidates are retained only when they include verbatim source spans, a source-document identifier, and a source URL. Under-specified or non-economic candidates are discarded. Generation prompts require self-contained questions with explicit country and temporal anchors where available, and a stratified subset is reserved for human auditing against the source documents. This process makes each retained instance traceable to evidence and supports direct inspection of benchmark quality.

We evaluate Qwen 3.6 35B, DeepSeek v4-pro, and Gemma 4 12B IT in three controlled conditions. Zero-shot provides no supporting context, oracle provides the short gold evidence passage, and RAG provides the top five chunks from a hybrid BM25 and BGE-M3 retriever fused using Reciprocal Rank Fusion. Together, these conditions contrast parametric recall, evidence use under gold context, and retrieval-augmented answering. Across models, zero-shot performance is low and retrieval substantially improves answer quality, yet list extraction, synthesis, and temporally scoped comparison remain difficult. AfriEconQA therefore provides both a benchmark and a controlled evaluation setup for studying where grounded QA systems fail in long, statistics-heavy institutional documents.

Our contributions are fourfold. First, we introduce **AfriEconQA**, comprising 4,309 evidence-linked QA instances from 220 economic reports on African economies. Second, we define a five-category taxonomy spanning precise extraction, structured recovery, synthesis, and comparison. Third, our evaluation separates parametric recall, evidence use, and retrieval. Finally, we analyze evidence-localization and generation failures on the held-out test split.

2 Related Work

2.1 Document-Grounded QA and Retrieval

Extractive benchmarks such as SQuAD Rajpurkar et al. (2016) evaluate whether a model can recover an answer from a supplied passage. Open-domain benchmarks extend this setting to retrieval from broader collections, including questions with many answers Amouyal et al. (2023), cross-lingual retrieval Liu et al. (2019); Longpre et al. (2021), and historical news archives Wang et al. (2022). Retrieval-augmented generation (RAG) similarly conditions generation on evidence retrieved from an external corpus Lewis et al. (2020), and domain-adapted RAG has been studied as a means of incorporating specialized knowledge Siriwardhana et al. (2023). AfriEconQA adopts this retrieve-then-generate setting, but uses it to isolate failure sources: zero-shot, oracle, and RAG conditions separate parametric recall, evidence use, and evidence retrieval. The relevant question is therefore not only whether a system retrieves a related document, but whether it localizes the passage that supports the required country, period, and projection state.

2.2 Financial Document Question Answering

Financial QA provides the closest methodological precedent because it combines text, tables, numerical reasoning, and source-grounded answers. FinQA Chen et al. (2021) contains 8,281 expert-authored questions over S&P 500 earnings reports and annotates executable reasoning programs. ConvFinQA Chen et al. (2022) extends this setting to multi-turn conver-

sational numerical reasoning over the same domain. AfriEconQA instead targets single-turn, evidence-linked questions over country-level economic reports. DocFinQA Reddy et al. (2024) places 7,437 FinQA questions back into their full SEC filings, increasing the average context from fewer than 700 words to approximately 123,000 words and directly evaluating retrieval over long documents. FinanceBench Islam et al. (2023) contributes 10,231 questions about publicly traded companies with corresponding answers and evidence strings, and evaluates retrieval and long-context configurations. Together, these benchmarks establish numerical reasoning, evidence grounding, and long-document retrieval as central problems in financial QA.

Two further resources broaden this setting. FinTextQA Chen et al. (2024a) contains 1,262 source-attributed, long-form QA pairs drawn from finance textbooks and government websites, with paragraph-length answers and RAG baselines. SEC-QA Lai et al. (2025) introduces a refreshable framework for generating quantitative, multi-document questions from financial metrics and associated filings, and reports document- and page-level retrieval evaluation. These resources are directly relevant to AfriEconQA's emphasis on long contexts and auditable evidence. Their primary tasks, however, concern company finance, financial regulation, or general financial knowledge rather than country-level development reporting with repeated country/time/projection-state near misses. To our knowledge, no prior QA benchmark packages World Bank country economic updates as an evidence-linked RAG test.

2.3 Economic Reasoning and Evidence Integration

Reasoning over economic information can require preserving relationships among several claims rather than extracting a single local fact. EconLogicQA Quan & Liu (2024) isolates one form of this problem by asking models to order logically connected events in multiple-choice scenarios generated from business news. It does not evaluate retrieval from a fixed institutional corpus. MMQA Wu et al. (2025) examines evidence integration in a different representation, requiring retrieval and inference across linked structured tables. AfriEconQA complements these tasks through Synthesis and Comparison questions over unstructured policy prose, where the relevant relations are expressed across claims, periods, and projection states rather than event-order options or relational tables.

AfriEconQA therefore targets a specific unresolved setting: evidence-linked QA over long institutional development reports, instantiated on a scoped World Bank Africa corpus. The reports combine fiscal statistics, policy conditions, historical outcomes, current estimates, and projections, often repeating the same indicator across countries and periods. The benchmark evaluates whether systems can distinguish these near-matching claims and preserve their numerical and temporal scope. This focus complements prior financial and economic benchmarks without treating them as instances of the same document or reasoning task.

3 Dataset

AfriEconQA is an evidence-linked question-answering benchmark over World Bank economic reports focused on African economies. Each instance contains a self-contained question q, gold answer a, short evidence field e, verbatim source spans r (`raw_evidence`), and document provenance d through a source-file identifier and URL. Multiple Choice instances additionally include four answer options.

Table 1 shows a representative instance. A nearby statistic for the wrong year, country, or projection state can appear plausible. Correctness therefore requires exact grounding rather than topical overlap.

3.1 Dataset Sources

The AfriEconQA corpus comprises **220** World Bank economic reports from **2024** and **2025** that focus on African economies World Bank. We use this regional boundary to make

Field	Example
Type	Comparison
Question	What is the projected change in monetary poverty in Burkina Faso between 2023, 2024, and 2025?
Answer	A decrease of 2.6 percentage points between 2023 and 2024, and another 3.3 percentage points by 2025
Evidence	Monetary poverty (at the international poverty line) in Burkina Faso is projected to have fallen 2.6 percentage points between 2023 and 2024 and to fall another 3.3 percentage points by 2025
Source	Burkina Faso Poverty and Equity Brief, October 2025

Table 1: Example AfriEconQA instance (id 1995) requiring country-scoped, multi-year, and projection-sensitive grounding.

systematic collection and auditing feasible, not because the underlying grounding problem is unique to Africa. The benchmark addresses a broader challenge in institutional reporting: linking an answer to the correct country, reporting period, and projection status.

The collection spans economic updates, poverty and equity briefs, policy notes, sectoral diagnostics, and development reports covering growth, debt, trade, inflation, labor markets, public finance, and related policy conditions. It also contains a small number of regional or MENA reports that follow the same reporting conventions. Every instance retains a source URL, while the original PDFs are not redistributed.

These documents create two challenges for document-grounded QA. First, temporal ambiguity is pervasive: country update series are republished over time, and a single report may place historical outcomes, current estimates, and forecasts for the same indicator in close proximity. Second, answers are highly specific. Many gold answers are exact percentages, fiscal years, monetary values, program names, or enumerated policy factors. Topical retrieval is therefore insufficient if the selected passage attaches the right indicator to the wrong country, period, or projection state.

3.2 Dataset Construction

Dataset construction proceeds in two stages: agentic candidate generation and automatic evidence grounding. An open-weight generator (`Llama-3.3-70B-Instruct`) produces candidate question–answer pairs from long document chunks through five category-specific agents:

- **Factoid:** exact numerical, temporal, or categorical claims.
- **List:** multiple factors, risks, or interventions stated in the evidence.
- **Multiple Choice:** four-option discrimination with distractors from the same economic context.
- **Synthesis:** integration of connected policy claims or economic mechanisms.
- **Comparison:** contrasts across periods, countries, indicators, or projected versus observed values.

Generation prompts require self-contained questions with explicit country and temporal anchors whenever available in the source text, and discourage meta-language such as "according to the report." Full generation, evaluation, and judge prompt templates are in Appendix C. The short evidence field e may be a direct quote or a concise synthesis for readability. The hard grounding constraint is on `raw_evidence`: every retained instance must include one or more verbatim source spans from the generation chunk, together with a source-document identifier and source URL. Candidates are discarded when `raw_evidence` is not a verbatim substring of the source chunk, when the question is under-specified, or when the content is not substantively economic. A stratified subset of 200 examples (40 per question type) is reserved for human auditing of gold-label quality against the source documents (Section 6).

3.3 Dataset Statistics

AfriEconQA contains **4,309** evidence-linked instances over 220 source reports. Mean question, answer, and short-evidence lengths are 14.3, 13.2, and 29.0 words, respectively. The associated `raw_evidence` text averages 935.9 words per instance. For retrieval experiments, the corpus is indexed into 60,629 chunks of 1,000 characters with 200-character overlap, distinct from the longer generation chunks used during dataset construction. Table 2 reports the distribution across question types. For public release, we define deterministic, type-stratified 70/10/20 train/validation/test splits (3,016 / 431 / 862, seed 42). Primary experiments in this paper evaluate the held-out test split ($n = 862$). Full-benchmark results ($n = 4{,}309$) appear in Appendix A and show the same ranking trends.

Question Type	Count	%
Factoid	1,093	25.4
Multiple Choice	944	21.9
List	879	20.4
Synthesis	710	16.5
Comparison	683	15.8
Total	**4,309**	**100.0**

Table 2: Question-type distribution in AfriEconQA.

Table 3 situates AfriEconQA relative to the closest financial and economic QA resources. Prior benchmarks primarily evaluate company filings, earnings reports, textbooks, or business-news reasoning. AfriEconQA differs in source domain and evaluation target: institutional development reports on African economies, with evidence-linked questions that require country-, period-, and projection-sensitive grounding.

Dataset	Primary source	#QA	#Docs	Evidence	Focus
FinQA	S&P 500 earnings reports	8,281	–	Program + context	Numerical reasoning
ConvFinQA	S&P 500 earnings reports	14,115	–	Conversation + context	Multi-turn numerical QA
DocFinQA	Full SEC filings	7,437	801	Full document	Long-context financial QA
FinanceBench	Public company filings	10,231	–	Evidence strings	Open-book financial QA
FinTextQA	Finance textbooks / agency sites	1,262	80	Document context	Long-form financial QA
EconLogicQA	Business news	650	–	MC event order	Economic sequential reasoning
AfriEconQA	World Bank African economic reports	**4,309**	**220**	Evidence + URL	Quantitative / temporal grounding

Table 3: Comparison of AfriEconQA with related financial and economic QA datasets. Document counts are reported only when stated by the source paper.

4 Experimental Setup

We evaluate contemporary LLMs on the held-out test split ($n = 862$) without fine-tuning on the released train split, which remains available for future supervised and transfer studies. Rather than proposing a new RAG architecture, the experiments vary the available evidence under controlled conditions to identify where strong generators fail on long-form economic-report QA.

4.1 Generator Models

We evaluate the following three generators. For brevity, subsequent tables refer to them as Qwen, DeepSeek, and Gemma.

- **Qwen 3.6 35B**, evaluated locally through LM Studio on Apple Silicon (48 GB unified memory).
- **DeepSeek v4-pro** (`deepseek-v4-pro`), evaluated through the DeepSeek API.

- **Gemma 4 12B IT** (`google/gemma-4-12b`), evaluated locally through LM Studio on the same host.

4.2 Experimental Conditions

Each generator model is evaluated across three conditions that contrast distinct sources of success and failure:

- **Zero-Shot:** The model receives only the question, with no retrieved context. This condition measures whether the answer is available through parametric memory.
- **Oracle:** The model receives the question and the short gold evidence passage e. This removes retrieval error and tests evidence use. Oracle uses one passage, whereas RAG uses five chunks.
- **Retrieval-Augmented Generation (RAG):** The model receives retrieved evidence from a hybrid BM25 Whissell & Clarke (2011) and BGE-M3 Chen et al. (2024b) retriever combined with Reciprocal Rank Fusion (RRF) Cormack et al. (2009), using the top $k = 5$ chunks per question.

For Multiple Choice questions, the prompt includes the answer choices in lettered form (A–D), and the model is instructed to return the option letter with a short answer phrase. This keeps the MCQ condition faithful to the released benchmark format and avoids penalizing models for not seeing the candidate options.

4.3 Answer Metrics

We use deterministic answer metrics because small numerical or temporal errors change the meaning of economic answers. Let $\hat{a}_i$ be the model prediction and a_i the gold answer for question i. We report 95% bootstrap confidence intervals for EM and F1 ($n = 1000$).

- **Exact Match (EM):** fraction of exact matches after normalization,

$$\text{EM} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\text{norm}(\hat{a}_i) = \text{norm}(a_i)].$$

  Numerical Factoid/Comparison answers allow $\pm 1\%$ tolerance (e.g., 78% vs. 78.2%). Multiple Choice scores the extracted option letter (A–D).

- **Token F1:** harmonic mean of token precision and recall,

$$\text{F1} = \frac{2 \cdot P \cdot R}{P + R}, \quad P = \frac{|\hat{T} \cap T|}{|\hat{T}|}, \quad R = \frac{|\hat{T} \cap T|}{|T|},$$

  where $\hat{T}$ and T are prediction and gold token sets. List questions use Set F1 over answer items so partial set recovery is credited.

- **ROUGE-L** Lin (2004): longest common subsequence (LCS) overlap, useful for longer Synthesis and Comparison answers,

$$\text{ROUGE}{-}\text{L} = \frac{(1 + \beta^2) R_{\text{lcs}} P_{\text{lcs}}}{R_{\text{lcs}} + \beta^2 P_{\text{lcs}}},$$

  with $R_{\text{lcs}} = \text{LCS}/|T|$, $P_{\text{lcs}} = \text{LCS}/|\hat{T}|$, and $\beta = 1$.

- **LLM Judge** Zheng et al. (2023): mean correctness score in $[0, 1]$ from `Llama-3.3-70B-Instruct` via the Groq API, used as a secondary semantic check when answers are paraphrased but still correct. Because the same model family also generated the dataset, we treat LLM-judge as supportive only. Main claims use EM and F1.

4.4 Retrieval Metrics

Let G_i be the gold evidence-supporting chunks for question i, and let $R_i^k = [r_{i1}, \ldots, r_{ik}]$ be the top-k retrieved chunks. Gold chunks are aligned to the retrieval index by evidence and answer spans.

- **HR@1:** whether the top retrieved chunk is gold,

$$\text{HR@1} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[r_{i1} \in G_i].$$

- **HR@5:** whether any gold chunk appears in the top five,

$$\text{HR@5} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\exists j \leq 5 : r_{ij} \in G_i].$$

- **MRR@5:** reciprocal rank of the first gold chunk in the top five (0 if none),

$$\text{MRR@5} = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{\text{rank}_i}.$$

- **Doc@1 / Doc@5 / Doc MRR@5:** same formulas as above, but a hit counts when any retrieved chunk comes from the correct source document rather than the exact evidence chunk.

5 Results

Table 4 gives the central benchmark result. Across all three model families, performance is low without evidence and rises sharply when source context is supplied. The LLM judge shows the same pattern, so the gap is not only an artifact of strict lexical matching. At the same time, the best RAG result still leaves substantial headroom, indicating that AfriEconQA is not solved by simply attaching a standard retriever to a strong generator.

Model	Condition	EM	F1	ROUGE-L	LLM-judge
Qwen	Zero-shot	0.165 [0.140, 0.190]	0.228 [0.204, 0.252]	0.170	0.415
	Oracle	0.390 [0.355, 0.422]	0.489 [0.456, 0.522]	0.279	0.595
	RAG	0.419 [0.384, 0.452]	0.525 [0.494, 0.556]	0.305	0.701
DeepSeek	Zero-shot	0.202 [0.173, 0.230]	0.267 [0.240, 0.294]	0.193	0.488
	Oracle	0.378 [0.345, 0.411]	0.473 [0.441, 0.504]	0.272	0.564
	RAG	0.434 [0.400, 0.468]	0.545 [0.516, 0.576]	0.315	0.723
Gemma	Zero-shot	0.162 [0.138, 0.188]	0.223 [0.199, 0.248]	0.167	0.406
	Oracle	0.319 [0.285, 0.350]	0.403 [0.372, 0.433]	0.237	0.497
	RAG	0.407 [0.375, 0.440]	0.509 [0.480, 0.541]	0.297	0.698

Table 4: Main results on the AfriEconQA test split ($n = 862$). All metrics in $[0, 1]$. EM and F1 include 95% bootstrap CIs. LLM-judge is the mean Groq-hosted Llama-3.3-70B LLM-as-a-judge score (secondary). Full-benchmark results are in Appendix A.

5.1 The Parametric Vacuum

The zero-shot condition measures whether answers can be recovered from parametric knowledge or generic economic priors. They generally cannot. Even after correcting the Multiple Choice setting so that answer options are visible and scored as letter selection, aggregate zero-shot F1 remains only 0.223–0.267, with exact match at or below 0.202 for every model. This is the expected behavior for a benchmark whose answers are tied to exact fiscal periods, projections, country-specific program names, and recently published report claims. In other words, AfriEconQA does not reward broad familiarity with economic development discourse. It requires access to the source document.

5.2 Evidence Grounding

Adding evidence changes model behavior substantially. RAG improves over zero-shot for every model, with DeepSeek obtaining the strongest aggregate RAG score (0.434 EM and 0.545 F1), followed closely by Qwen and Gemma. The oracle condition also improves sharply over zero-shot, confirming that many errors are evidence-access errors rather than pure generator incapability.

The relationship between oracle and RAG is meant for comparison rather than as a simple upper-bound claim. The oracle setting provides the model with the short gold evidence passage, while RAG provides the top five retrieved chunks. Aggregate RAG F1 exceeds single-passage oracle F1 for all three model families, suggesting that some AfriEconQA questions benefit from multi-chunk context even when a gold passage is available. We therefore use the oracle condition to test whether a model can use relevant evidence, not as a claim that single-passage oracle evidence is the maximum achievable context.

5.3 Retrieval Is Mostly a Localization Problem

The aggregate RAG results leave open an important question: does retrieval fail because systems retrieve the wrong report, or because they retrieve the right report but miss the exact claim needed to answer the question? We answer this with a retriever-only analysis on the test split, comparing BM25, BGE-M3, and the Hybrid/RRF retriever used in the RAG condition. Gold chunks are resolved by matching evidence/answer spans into the 1,000-character retrieval index. When no span match is found, we fall back to the instance's labeled generation `chunk_id` if that chunk exists for the source document (**17**/862 test questions, 2.0%).

Metric	BM25	BGE-M3	Hybrid
HR@1	0.275	0.309	0.328
HR@5	0.423	0.492	0.502
MRR@5	0.330	0.382	0.394
Doc@1	0.759	0.797	0.841
Doc@5	0.914	0.945	0.961
Doc MRR@5	0.819	0.859	0.888

Table 5: Retriever-only results on the test split ($n = 862$). All metrics are in $[0, 1]$. Strict metrics require an evidence-supporting chunk, whereas Doc@* only require the correct source document.

Table 5 shows a sharp separation between document retrieval and evidence localization. Document-level HR@5 is high (0.914–0.961), so the correct World Bank report is usually retrieved. Strict chunk-level HR@5 is much lower (0.423–0.502), meaning that the exact evidence-bearing chunk is often absent from the five chunks passed to the generator. This is a central empirical property of AfriEconQA. The benchmark does not merely ask whether a retriever can identify a topically related report. It asks whether the system can isolate the specific country-, year-, indicator-, or policy-scoped claim inside that report.

BGE-M3 improves over BM25 on all strict retrieval metrics, showing that dense semantic matching helps beyond lexical overlap. Hybrid/RRF gives the best HR@1 and MRR@5, indicating that lexical and semantic signals are complementary for ranking. However, Hybrid/RRF improves HR@5 only marginally over BGE-M3 (0.502 vs. 0.492). The right conclusion is therefore not that hybrid retrieval solves the task, but that even a reasonable hybrid retriever often localizes the document better than the evidence. This explains why RAG helps substantially while still leaving large residual error.

5.4 Open-Ended Reasoning Remains Hard

Table 6 shows that aggregate scores hide large type-level differences. Multiple Choice is comparatively easy once options are visible (RAG F1 above 0.78 for all models). Factoid

questions benefit strongly from retrieval, with RAG F1 above 0.69 across models. List questions remain hard even with gold evidence (oracle F1 0.14–0.18), and RAG F1 stays near 0.08–0.10, indicating that retrieved near-neighbor chunks often interfere with complete set recovery. Synthesis and Comparison sit in between: retrieval helps, but temporal scope and structured comparison remain difficult.

Model	Type	ZS	Oracle	RAG
Qwen	Comparison	0.116	0.361	0.474
	Factoid	0.033	0.580	0.693
	List	0.008	0.165	0.084
	MC	0.704	0.910	0.857
	Synthesis	0.271	0.309	0.416
DeepSeek	Comparison	0.132	0.361	0.508
	Factoid	0.112	0.515	0.705
	List	0.010	0.180	0.101
	MC	0.773	0.921	0.884
	Synthesis	0.278	0.281	0.433
Gemma	Comparison	0.097	0.296	0.466
	Factoid	0.030	0.464	0.713
	List	0.007	0.142	0.081
	MC	0.709	0.778	0.783
	Synthesis	0.261	0.236	0.399

Table 6: Test-split F1 by question type (3 d.p., $[0, 1]$) under zero-shot (ZS), oracle, and RAG ($n = 862$).

6 Human Evaluation

AfriEconQA includes a stratified human-evaluation subset of 200 examples sampled from the full 4,309-instance pool (not from a single split), with 40 examples from each question type (seed 42). This evaluation audits *benchmark quality*, not model outputs. Independent evaluators inspect each item against the source PDF and label answer validity as `correct`, `partial`, `incorrect`, or `unanswerable`. Three independent evaluators each reviewed all 200 items, producing 600 labels. Agreement remained stable as annotation expanded. The first pair agreed on **89%** of items, with Gwet's AC1 at **0.85**. Across all three annotator pairs, observed agreement was **88%** (530 of 600 pairwise comparisons), with an AC1 of **0.88**. All three evaluators assigned the same label to **83%** of items (166/200). We report AC1 because it is less sensitive than κ-type statistics to the strong prevalence of `correct` labels Feinstein & Cicchetti (1990).

Across the 600 individual judgments, **92%** were `correct`. Majority voting resolved 198 of 200 items, and **95%** of those majorities were `correct` (189/198). The 95% Wilson confidence interval is $[92\%, 98\%]$. Majority-correct rates ranged from 92% to 98% across question types. This evaluation audits whether gold answers are supported by the source PDF rather than estimating human performance on the QA task.

6.1 Dataset Explorer

To make the benchmark auditable during review, we include screenshots of the dataset explorer used for sample inspection and human audit (Figures 1 and 2). The interface exposes the question, gold answer, supporting evidence, source-document metadata, and the corresponding location in the source PDF where available. It is the same interface used for the stratified human audit, so reviewers can see how gold-label quality is checked against the source document. This is not a separate model-evaluation task. It is a transparency mechanism that lets reviewers verify the benchmark's grounding assumptions rather than treating the dataset as an opaque file.

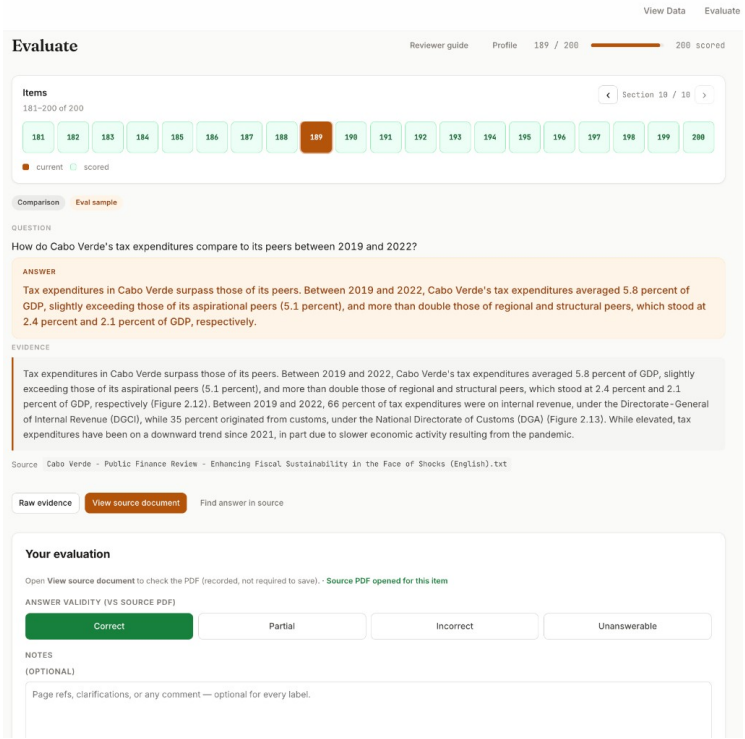


Figure 1: AfriEconQA explorer audit view: question, gold answer, evidence, and validity labels.



Figure 2: Source-PDF view in the explorer, with the supporting evidence span highlighted in the World Bank report.

7 Error Analysis

Two failure modes dominate. First, retrieval often finds the right report but not the exact claim: document-level HR@5 is high, while chunk-level HR@5 remains near 0.5 (Table 5). World Bank reports reuse the same indicators across countries, editions, and horizons, so topically related chunks are frequent near misses. Second, generation remains hard when the answer must be structured. List questions stay weak under both oracle and RAG, so missing evidence alone does not explain the errors. Models also fail at complete set recovery. Together with Table 6, these patterns show that AfriEconQA stresses evidence localization and structured answer generation more than topical document retrieval.

8 Conclusion

We introduced AfriEconQA, a benchmark of 4,309 evidence-linked questions for document-grounded QA over World Bank economic reports. The benchmark targets quantitative, temporal, and structural reasoning over institutional documents whose claims are difficult to answer from parametric memory alone. Across three model families and three controlled conditions, zero-shot performance remains low, RAG substantially improves answer quality, and type-level analysis reveals persistent weaknesses on List, Synthesis, and Comparison questions.

These findings show that reliable economic-report QA requires more than topical retrieval. Systems must retrieve the correct country- and time-scoped evidence, preserve exact numerical values, and transform dense policy prose into structured answers. AfriEconQA provides an evidence-grounded and auditable testbed for measuring progress on these capabilities.

9 Reproducibility and Licensing

AfriEconQA is released under a Creative Commons Attribution 4.0 (CC BY 4.0) license. Source World Bank reports remain under World Bank Open Access / CC BY 4.0 terms. Each instance retains a source URL for attribution, and the original PDFs are not redistributed. The release includes the evidence-linked QA file, deterministic type-stratified train/validation/test splits (seed 42, with primary evaluation on test, $n = 862$), the stratified 200-item human-audit subset, and evaluation scripts under an open-source license. Prompts are in Appendix C; fixed implementation settings are in Appendix B and `config.py` in the supplementary Code and Data package.

10 Limitations

AfriEconQA is scoped to World Bank economic reports on African economies from 2024–2025, so results should not be read as covering all economic document QA. Gold answers reflect the cited report edition, while tracking later revisions is out of scope. Human validation audits gold-label support against the source PDF on a stratified 200-item subset. It does not measure human performance on the full open-ended QA task, which would require different expertise and protocol. The LLM judge is a secondary semantic signal and can be more lenient than exact match on long answers. Oracle uses a single gold evidence passage while RAG uses top-five chunks, so oracle is a controlled evidence-use condition rather than a strict performance ceiling. Stronger specialized retrievers and longer-context generators may further improve absolute scores. Our contribution is the benchmark and the controlled evaluation protocol.

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A Full-Benchmark Results

Primary results in the main paper use the held-out test split ($n = 862$). For completeness, Tables 7–9 report the same metrics on the full benchmark ($n = 4{,}309$). Relative model rankings and the document-vs-chunk localization gap are unchanged.

Model	Condition	EM	F1	ROUGE-L	LLM-judge
Qwen	Zero-shot	0.162 [0.151, 0.173]	0.228 [0.218, 0.239]	0.174	0.411
	Oracle	0.392 [0.377, 0.408]	0.494 [0.480, 0.508]	0.286	0.605
	RAG	0.426 [0.411, 0.440]	0.537 [0.524, 0.550]	0.314	0.708
DeepSeek	Zero-shot	0.190 [0.178, 0.201]	0.259 [0.247, 0.270]	0.192	0.477
	Oracle	0.375 [0.360, 0.389]	0.474 [0.461, 0.488]	0.279	0.572
	RAG	0.439 [0.424, 0.453]	0.551 [0.537, 0.563]	0.319	0.721
Gemma	Zero-shot	0.159 [0.148, 0.170]	0.224 [0.213, 0.234]	0.169	0.404
	Oracle	0.327 [0.312, 0.340]	0.416 [0.403, 0.430]	0.252	0.517
	RAG	0.415 [0.400, 0.429]	0.523 [0.511, 0.537]	0.307	0.702

Table 7: Full-benchmark main results ($n = 4{,}309$). All metrics in $[0, 1]$. EM and F1 include 95% bootstrap CIs. LLM-judge is the mean Groq-hosted Llama-3.3-70B LLM-as-a-judge score (secondary).

Metric	BM25	BGE-M3	Hybrid
HR@1	0.271	0.307	0.325
HR@5	0.432	0.490	0.496
MRR@5	0.333	0.379	0.392
Doc@1	0.741	0.805	0.825
Doc@5	0.902	0.945	0.954
Doc MRR@5	0.806	0.864	0.879

Table 8: Full-benchmark retriever-only results ($n = 4{,}309$). All metrics in $[0, 1]$.

Model	Type	ZS	Oracle	RAG
Qwen	Comparison	0.129	0.369	0.486
	Factoid	0.050	0.579	0.705
	List	0.006	0.154	0.084
	MC	0.677	0.926	0.876
	Synthesis	0.276	0.332	0.441
DeepSeek	Comparison	0.161	0.342	0.507
	Factoid	0.118	0.500	0.726
	List	0.007	0.165	0.085
	MC	0.706	0.940	0.892
	Synthesis	0.288	0.324	0.445
Gemma	Comparison	0.117	0.297	0.490
	Factoid	0.042	0.474	0.708
	List	0.004	0.140	0.077
	MC	0.679	0.793	0.815
	Synthesis	0.270	0.282	0.436

Table 9: Full-benchmark F1 by question type ($n = 4{,}309$, 3 d.p., $[0, 1]$).

B Implementation Details

Table 10 lists the fixed settings used for the reported experiments. Each generator is evaluated once per condition on the held-out test split ($n = 862$). We did not tune hyperparame-

ters on the test split. The supplementary Code and Data package ships the same values in `config.py`.

Setting	Value
Generators	
Qwen 3.6 35B	`qwen/qwen3.6-35b-a3b` (LM Studio)
Gemma 4 12B IT	`google/gemma-4-12b` (LM Studio)
DeepSeek v4-pro	`deepseek-v4-pro` (DeepSeek API)
Decoding (all generators)	temperature 0.2, top-p 0.95, max 320 tokens
Evaluation runs	one pass per generator $\times$ condition
Retrieval (RAG)	
Retriever	BM25 + BGE-M3 (`BAAI/bge-m3`)
Fusion	Reciprocal Rank Fusion, c=60
Candidate depth / top-k	60 / 5
Index chunk size	1,000 characters, 200 overlap
Metrics and splits	
LLM judge	`Llama-3.3-70B-Instruct` (Groq), temp. 0, max 384 tokens
Bootstrap CIs (EM/F1)	1,000 resamples, seed 42
Train/val/test splits	seed 42; 3,016 / 431 / 862

Table 10: Fixed implementation settings (also in supplementary `config.py`).

C Prompts

We list the main prompts used for dataset generation, model evaluation, and LLM judging. Minor wording may vary by question type. The templates below are the ones used for the reported experiments.

C.0.1 QA Generation

Questions are produced by five type-specific agents (`Llama-3.3-70B-Instruct`). Each agent uses a system prompt with a role line, shared critical rules, and a JSON output schema, plus a user message that provides the report chunk.

Agent Role Lines

Factoid

You are a World Bank Data Agent. Extract Factoid QA pairs with precise numerical or categorical facts. Only generate if the text has strong, unambiguous statistical claims.

List

You are a World Bank Synthesis Agent. Extract List QA pairs identifying multiple discrete factors, risks, or policy drivers. Only generate if the text explicitly enumerates multiple items.

Comparison

You are a World Bank Analytical Agent. Extract Comparison QA pairs reasoning across time periods or regions. Only generate if the text explicitly compares data across years or countries.

Multiple Choice

You are a World Bank Assessment Agent. Generate Multiple Choice QA pairs with 4 options (A/B/C/D). Only generate if the text supports strong distractor construction.

Synthesis

You are a World Bank Macroeconomic Agent. Extract Synthesis QA pairs covering complex causal chains and economic transmission mechanisms. Only generate if the text explicitly describes how policy X caused outcome Y.

Shared Critical Rules (System)

1. DO NOT force questions. If the text chunk does not support good questions for this task, return `[]`.
2. Quality over quantity. Some chunks can produce 0 questions.
3. Questions must be self-contained and specific (name country/region/time period whenever available in the text).
4. DO NOT use meta-language like "According to the report" or "The document states".
5. Use semantic understanding: question, answer, and `evidence` may be paraphrased.
6. If one sentence in the TEXT CHUNK fully supports the answer and contains enough context, copy that sentence exactly into `evidence`.
7. If the answer requires combining multiple sentences, figures, table rows, or context from different parts of the chunk, write a concise synthesized evidence paragraph in `evidence`.
8. Whenever `evidence` is synthesized, include the original verbatim supporting sentences or table rows in `raw_evidence`.
9. `raw_evidence` must always contain only verbatim text copied from the TEXT CHUNK.
10. Do not require the answer wording to exactly match the evidence wording. Semantically equivalent wording is acceptable if grounded in `raw_evidence`.
11. For table-based questions, include row/value text. Do not use only the table title.
12. `evidence_rationale` must be exactly one sentence and 40 words or fewer.

Standard JSON Output Schema (non-MCQ)

```
Output strictly as a raw JSON array. No markdown, no explanation text before or after.

[
  {
    question_type: {q_type},
    question: ...,
    answer: ...,
    evidence: [Either one exact supporting sentence OR a synthesized
      evidence paragraph grounded in raw_evidence],
    raw_evidence: [verbatim supporting sentence/table row copied
      from TEXT CHUNK],
    evidence_rationale: Exactly 1 sentence and 40 words or less
      explaining why this evidence supports the answer.
  }
]
```

Multiple Choice Extras

Appended to the 12 shared rules above for the Multiple Choice agent only, replacing the standard output schema.

```
10. MULTIPLE CHOICE FORMAT: The output JSON for each MC item MUST include an
options field as a dictionary with exactly four keys: A, B, C, D.
Exactly one option is the correct answer. The other three are STRONG
DISTRACTORS -- they must be wrong, but plausible to someone who has not read
the report carefully (e.g., similar numbers, related-but-wrong institutions,
plausible-but-incorrect causal chains). Avoid obviously absurd distractors.

11. Do NOT always place the correct answer in option A. Randomize the
correct option position across questions.

Output format for Multiple Choice:
[
  {
    question_type: Multiple Choice,
```

```
    question: ...,
    answer: B,
    options: {A: ..., B: [CORRECT ANSWER TEXT], C: ...,
      D: ...},
    evidence: [direct sentence or concise synthesis...],
    raw_evidence: [exact source sentence/span from chunk...],
    evidence_rationale: One sentence, <= 40 words.
  }
]
```

User Message Template

REPORT FILENAME: ⟨filename⟩ (Part i/N)
TEXT CHUNK:
⟨chunk text⟩
Generate the ⟨Type⟩ JSON QA pairs now if applicable, otherwise return `[]`.

C.0.2 Evaluation Prompts

Zero-Shot Evaluation Prompt

Answer the following question about African macroeconomic policy accurately. Keep your answer concise and direct.

Question: ⟨question⟩
Answer:

(MCQ) Answer with the correct option letter (A, B, C, or D) followed by a short answer phrase. Do not include extra commentary.
(other) Answer in a single concise sentence with no extra commentary.

Oracle Evaluation Prompt

Using the provided context, answer the following question about African macroeconomic policy accurately. Keep your answer concise and direct. If the context does not contain the information needed to answer the question, output "Information not provided".

Context:
⟨gold evidence⟩

Question: ⟨question⟩
Answer: *(same answer-format instruction as zero-shot)*

RAG Evaluation Prompt

Using the provided retrieved documents, answer the following question about African macroeconomic policy accurately. Keep your answer concise and direct. If the documents do not contain the information needed to answer the question, output "Information not provided".

Documents:
⟨top-5 retrieved chunks⟩

Question: ⟨question⟩
Answer: *(same answer-format instruction as zero-shot)*

C.0.3 LLM Judge

The judge model is `Llama-3.3-70B-Instruct` on Groq, called once per prediction with a fixed system prompt and a per-item user prompt built from the template below.

Judge System Prompt

You are evaluating a generated answer against a reference answer for a factual QA benchmark. Score factual agreement with the reference only. Do not reward an answer for discussing the correct topic, sounding fluent, or being verbose. Reward it only for stating the same facts. A claim that contradicts the reference is worse than one it simply omits. Treat the reference as ground truth: do not use outside knowledge, and do not decide the reference itself is wrong. Output JSON only.

Per-Type Verification Instruction

One line is substituted for {`task_prompt`} in the user prompt below, selected by question type.

- **factoid:** Verify that the answer gives the correct entity or value, including any essential unit and time qualifier.
- **multiple_choice:** Verify that the answer selects the correct option. Compare the generated answer to the gold option TEXT exactly. If it states a value or phrase that matches a different option, score 0 even if the topic is right.
- **list:** Verify that the answer includes the required items. Give partial credit when only some required items are correct.
- **synthesis:** Verify the main conclusion and supporting relationship or mechanism. Topic overlap alone is not enough.
- **comparison:** Verify the compared entities, direction of the comparison, and any stated values or magnitudes.
- **default:** Verify that the generated answer correctly and completely answers the question.

Judge User Prompt Template

```
{task_prompt}

Question: {question}
Reference Answer: {truth}
Generated Answer: {prediction}
{options_block}

Score using this scale:
10 = fully correct and complete, with no missing or incorrect facts
8-9 = correct, with only a minor imprecision in phrasing and all facts correct
6-7 = mostly correct, missing one relevant fact but stating nothing that
      contradicts the reference
4-5 = missing more than one relevant fact, but nothing stated contradicts
      the reference
2-3 = contains a factual contradiction, or is on-topic with no correct
      facts
0-1 = incorrect, contradictory throughout, irrelevant, empty, or a
      refusal

A contradicted fact always scores in the 0-3 range, even if other facts in
the same answer are correct -- never average a contradiction against
surrounding correct facts.

List only facts explicitly stated in the reference or generated answer for
matched/missing/incorrect below -- do not infer implications, and do not
add facts that appear in neither answer.

Return exactly {score: <integer 0-10>, matched: [...], missing:
[...], incorrect: [...], reasoning: <one sentence>}.
```

For Multiple Choice items, {`options_block`} additionally lists the four options and the gold correct option letter. Reported LLM-judge scores normalize the returned integer to $[0, 1]$ by dividing by 10.